

Topic:-

Collect wrappers of ready-made food items. Prepare a chart of foodstuff and identify ingredients as organic or inorganic.

Introduction:-

In our daily lives, we eat many packaged and ready-made food items like biscuits, potato chips, noodles, milk chocolates, and fruit juices. Whenever we purchase packaged snacks from grocery stores, we see printed nutritional ingredient lists on the back of the wrappers. In science, all substances that make up matter are chemical compounds.

In Chapter 6 of our Class 8 Science textbook, chemical compounds are classified into two major categories: Organic compounds and Inorganic compounds. Organic compounds are carbon-containing compounds derived directly or indirectly from living plants and animals (such as sugars, starches, fats, and proteins). Inorganic compounds come from non-living mineral sources (such as common salt, water, and baking soda). In this project, we collected food wrappers and analyzed their ingredients.

❖ Main Differences: Organic vs Inorganic Compounds:-

- **Organic:** Contain carbon-hydrogen bonds; char into black carbon when heated.
- **Inorganic:** Mineral-based compounds without carbon chains; do not char easily.
- **Test:** Burning sugar turns black (organic); heating salt remains white (inorganic).

Aim & Survey Preparation:-

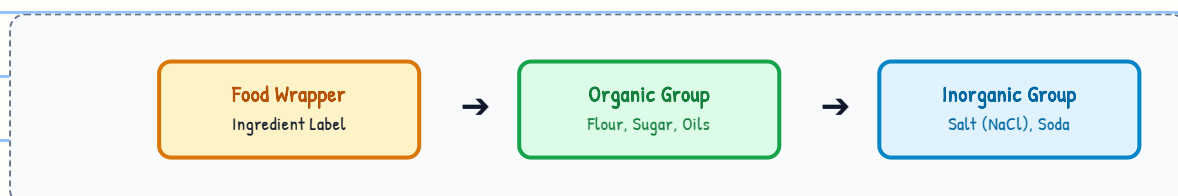
Aim: To collect discarded wrappers of ready-made food items, analyze their ingredient labels, prepare a comprehensive chart, and classify each ingredient as organic or inorganic.

❖ Objectives of the Project:-

- To read and understand nutritional labels printed on packaged snacks.
- To identify organic ingredients (carbohydrates, fats, proteins) and inorganic additives (salts, minerals).
- To evaluate the health impact of artificial preservatives, excess sodium, and refined sugars.

❖ Materials Required:-

- Empty clean wrappers: potato chips, Marie biscuits, Maggi noodles, dairy milk chocolate, fruit juice tetrapack.
- Large white chart paper, scissors, glue stick, ruler, and colored marker pens.



Procedure & Label Classification Method:-

We systematically cataloged and classified the collected wrappers using these steps:

- **Step 1: Collection & Cleaning:** Collected 10 different clean empty wrappers from home and friends (biscuits, chips, noodles, chocolate, juice, ketchup).
- **Step 2: Locating the Ingredients Panel:** Turned to the back of each wrapper and carefully read the fine-print list marked '*Ingredients*'.
- **Step 3: Chemical Identification:** Checked each item against chemistry references:
 - Natural biological extracts (wheat flour, palm oil, cocoa powder, glucose, starch) → **Organic**.
 - Mineral salts, chemical leavening agents, and water (sodium chloride, baking soda, calcium carbonate, iron salts) → **Inorganic**.
- **Step 4: Mounting on Chart:** Glued each wrapper neatly on our chart alongside its ingredient breakdown table.



Observations & Ingredient Analysis Table:-

The table below displays our classification of ingredients from common ready-made foods:

Food Item	Key Ingredients Listed	Organic Components	Inorganic Components
Marie Biscuits	Wheat flour, sugar, palm oil, salt, baking soda (INS 500)	Wheat flour, sugar, vegetable oil	Common salt (NaCl), Sodium bicarbonate
Potato Chips	Potatoes, edible oil, salt, spices, citric acid	Potato starch, oil, chili spices	Common salt, Potassium chloride
Instant Noodles	Refined wheat flour, palm oil, mineral salt, INS 508	Refined flour (Maida), palm oil	Sodium carbonate, Potassium chloride
Milk Chocolate	Sugar, cocoa butter, milk solids, soy lecithin	Sugar, cocoa butter, milk fat, lecithin	Mineral ash, trace calcium salts
Fruit Juice	Water, apple juice concentrate, sugar, acidity regulator (330)	Fructose fruit sugar, sucrose	Water (H ₂ O), Citric acid salts

Scientific Analysis & Discussion:-

Our classification revealed important scientific insights into the food manufacturing industry:

- **Dominance of Organic Compounds:** Over 85% of all ingredients in human food are organic. Our bodies need organic macronutrients—carbohydrates for energy, proteins for muscle growth, and fats for cell membranes.
- **Vital Role of Inorganic Compounds:** Even though inorganic ingredients make up only 2% to 5% of food weight, they are essential. Common salt (sodium chloride) maintains fluid balance and nerve transmission, while baking soda (sodium bicarbonate) releases CO_2 gas to make biscuits light and crisp.
- **Food Additives (INS Codes):** International Numbering System (INS) numbers represent food regulators and preservatives. For example, INS 500 is sodium bicarbonate, and INS 330 is citric acid.

❖ High Sodium and Sugar Warning:-

Packaged chips and noodles contain high levels of inorganic sodium salt. Eating too much packaged food regularly can cause high blood pressure and kidney strain.

Everyday Applications & Consumer Awareness:-

Learning to read food wrappers empowers us to make healthy dietary choices:

- **Checking the Green / Brown Dot:** The green circle symbol indicates 100% vegetarian ingredients, while the brown symbol indicates non-vegetarian ingredients.
- **Checking Expiry & Best Before Dates:** Never consume packaged food past its expiry date because oils oxidize and turn rancid, producing toxic compounds.
- **Spotting Hidden Sugars:** Manufacturers often hide sugar under scientific organic names like maltodextrin, high-fructose corn syrup, dextrose, and invert syrup.

❖ Key Recommendations for School Students:-

- Choose fresh homemade snacks (poha, upma, boiled sprouts, seasonal fruits) over processed chips.
- Drink fresh coconut water or homemade lemonade instead of sugary bottled sodas.
- Always wash hands before eating to avoid ingesting dirty plastic residues.

What We Learned & Conclusion:-

- Matter in food is divided into plant/animal-derived organic compounds and mineral-based inorganic compounds.
- Organic compounds form the main caloric nutrients (starch, sugar, fats), while inorganic compounds act as salts and leavening agents.
- Reading food wrappers helps consumers detect hidden chemicals, excess salt, and synthetic preservatives.
- Balancing natural organic foods and limiting processed packaged snacks ensures lifelong health.

Conclusion:-

Preparing this foodstuff chart by collecting and analyzing real wrappers proved that chemistry is present in every bite we eat. We successfully classified common ingredients into organic and inorganic compounds and gained valuable consumer awareness that will help our families make healthier, informed food choices.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 6: Composition of Matter).
- Food Safety and Standards Authority of India (FSSAI) – Reading Food Labels Guide.
- Nutrition and Chemistry in Daily Life, Balbharti Pune.